12.3: The eukaryotic cell cycle is regulated by a molecular control system

The timing and rate of cell division in different parts of a plant or animal are crucial to normal growth, development, and maintenance. The frequency of cell division varies with the type of cell. For example, human skin cells divide frequently throughout life, whereas liver cells maintain the ability to divide but keep it in reserve until an appropriate need arises—say, to repair a wound. Some of the most specialized cells, such as fully formed nerve cells and muscle cells, do not divide at all in a mature human. These cell cycle differences result from regulation at the molecular level. The mechanisms of this regulation are of great interest, not only to understand the life cycles of normal cells but also to learn how cancer cells manage to escape the usual controls.

The Cell Cycle Control System

What controls the cell cycle? In the early 1970s, a variety of experiments led to the hypothesis that the cell cycle is driven by specific signaling molecules present in the cytoplasm. Some of the first strong evidence for this hypothesis came from experiments with mammalian cells grown in culture. In these experiments, two cells in different phases of the cell cycle were fused to form a single cell with two nuclei **(Figure 12.14)**. If one of the original cells was in the S phase and the other was in G1, the G1 nucleus immediately entered the S phase, as though stimulated by signaling molecules present in the cytoplasm of the first cell. Similarly, if a cell undergoing mitosis (M phase) was fused with another cell in any stage of its cell cycle, even G1, the second nucleus immediately entered mitosis, with condensation of the chromatin and formation of a mitotic spindle.

Figure 12.14

Inquiry: Do molecular signals in the cytoplasm regulate the cell cycle?

Experiment

Researchers at the University of Colorado wondered whether a cell's progression through the cell cycle is controlled by cytoplasmic molecules. They induced cultured mammalian cells at different phases of the cell

cycle to fuse. Two experiments are shown.

Results

Two diagrams of experiments involving cells at different phases of the cell cycle.

Experiment 1: A cell drawn in S-Phase is fused to a cell in G 1 phase. The arrow shows these cells go to one large, fused cell with two nuclei both in S-phase and the D N A is synthesized.

Experiment 2: One cell drawn in M fused with another cell in G 1. The arrow shows the cells go to one large, fused cell with two nuclei both in M; the mitotic spindle is visible. However, the chromosome that had been in G 1 has entered mitosis even though it is unduplicated.

Conclusion

The results of fusing a G1 cell with a cell in the S or M phase of the cell cycle suggest that molecules present in the cytoplasm during the S or M phase control the progression to those phases.

Data from R. T. Johnson and P. N. Rao, Mammalian cell fusion: Induction of premature chromosome condensation in interphase nuclei, *Nature*226:717–722 (1970).

WHAT IF?***If the progression of phases did not depend on cytoplasmic molecules and each phase began when the previous one was complete, how would the results have differed?*

*For suggested answer, see**Appendix A*.

The experiment shown in Figure 12.14 and other experiments on animal cells and yeasts demonstrated that the sequential events of the cell cycle are directed by a distinct **cell cycle control system**, a cyclically operating set of molecules in the cell that both triggers and coordinates key events in the cell cycle **(Figure 12.15)**. The cell cycle control system has been compared to the control device of a washing machine. Like the washer's timing device, the cell cycle control system proceeds on its own, according to a built-in clock. However, just as a washer's cycle is subject to both internal control (such as the sensor that detects when the tub is filled with water) and external adjustment (such as starting or stopping the machine), the cell cycle is regulated at certain checkpoints by both internal and external signals. A **checkpoint** in the cell cycle is a control point where stop and go-ahead signals can regulate the cycle. Three important checkpoints are found in the G1, G2, and M phases (red gates in Figure 12.15), which will be discussed shortly.

Figure 12.15

Mechanical analogy for the cell cycle control system.

Clock-like diagram of the cell cycle, with perimeter segregated into sections depicting sequential events.

In this diagram, the flat "stepping stones" around the perimeter represent sequential events. Like the control device of a washing machine, the cell cycle control system proceeds on its own, driven by a built-in clock. However, the system is subject to internal and external regulation at various checkpoints; three important checkpoints are shown (red).

Inner part of the clock is separated into pie-shaped slices representing G 1, S, G 2, and M phases. Arrows indicate a clockwise progression of these events. The cell's control system is separated by 3 checkpoints (red gates in the diagram): the G 1 checkpoint (located near the end of G 1), the G 2 checkpoint (located at the end of G 2), and the M checkpoint (located in the middle of the M slice).

Animation

Control of the Cell Cycle

image

Play

image

Play

To understand how cell cycle checkpoints work, we'll first identify some of the molecules that make up the cell cycle control system (the molecular basis for the cell cycle clock) and describe how a cell progresses through the cycle. We'll then consider the internal and external checkpoint signals that can make the clock either pause or continue.

The Cell Cycle Clock: Cyclins and Cyclin-Dependent Kinases

Rhythmic fluctuations in the abundance and activity of cell cycle control molecules pace the sequential events of the cell cycle. These regulatory molecules are mainly proteins of two types: protein kinases and cyclins. As you learned in Concept 11.3, protein kinases are enzymes that activate or inactivate other proteins by phosphorylating them.

Many of the kinases that drive the cell cycle are actually present at a constant concentration in the growing cell, but much of the time they are in an inactive form. To be active, such a kinase must be attached to a **cyclin**, a protein that gets its name from its cyclically fluctuating concentration in the cell. Because of this requirement, these kinases are called **cyclin-dependent kinases**, or **Cdks**. The activity of a Cdk rises and falls with changes in the concentration of its cyclin partner. **Figure 12.16a** shows the fluctuating activity of

MPF, the cyclin-Cdk complex that was discovered first (in frog eggs). Note that the peaks of MPF activity correspond to the peaks of cyclin concentration. The cyclin level rises during the S and G2 phases and then falls abruptly during M phase.

Figure 12.16

Molecular control of the cell cycle at the G2 checkpoint.

Molecular control of the cell cycle at the G 2 checkpoint.

Part a. Fluctuation of M P F activity and cyclin concentration during the cell cycle. The graph shows cyclin concentration and M P F activity build during S and G 2 phases and fall off sharply during the M phase. The graph shows 3 peaks for M P F activity and cyclin concentration (y-axis) over time (x-axis) during different phases of the cell cycle. During the first peak, cyclin concentration rises sharply before the M phase and then abruptly falls off during the M phase, while M P F activity has a symmetric (hump-shaped) increase and decrease around the same point. In the second peak, cyclin concentration rises sharply during the S and G 2 phases before falling abruptly during the M phase. M P F activity increases and decreases symmetrically before and during the M phase. The third peak follows the same pattern and peak 2.

Part b. Molecular mechanisms that help regulate the cell cycle.

- Synthesis of cyclin begins in the late S phase and continues through G 2. Because cyclin is protected from degradation during this stage, it accumulates.
- Cyclin combines with Cdk, producing M P F. When enough M P F molecules accumulate, the cell passes the G 2 checkpoint and begins mitosis.
- M P F promotes mitosis by phosphorylating various proteins. M P F‘s activity peaks during metaphase.
- During anaphase, the cyclin component of M P F is degraded, terminating the M phase. The cell enters the G 1 phase.
- 5. During G 1, the degradation of cyclin continues, and the Cdk component of M P F is recycled.

The steps of the cell cycle are timed by rhythmic fluctuations in the activity of cyclin-dependent kinases (Cdks). Here we focus on a cyclin-Cdk complex in animal cells called MPF, which acts at the G2 checkpoint as a go-ahead signal, triggering the events of mitosis.

VISUAL SKILLS *Explain how the events in the diagram in (b) are related to the “Time” axis of the graph in (a), beginning at the left.*

*For suggested answer, see*Appendix A.**

Interview

Interview with Paul Nurse: Discovering how protein kinases control the cell cycle

Paul Nurse.

The initials MPF stand for "maturation-promoting factor," but we can think of MPF as "M-phase-promoting factor" because it triggers the cell's passage into the M phase, past the G2 checkpoint. When cyclins that accumulate during G2 associate with Cdk molecules, the resulting MPF complex is active—it phosphorylates a variety of proteins, initiating mitosis **(Figure 12.16b)**. MPF acts both directly as a kinase and indirectly by activating other kinases. For example, MPF causes phosphorylation of various proteins of the nuclear lamina (see Figure 6.9), which promotes fragmentation of the nuclear envelope during prometaphase of mitosis. There is also evidence that MPF contributes to molecular events required for chromosome condensation and spindle formation during prophase.

During anaphase, MPF helps switch itself off by initiating a process that leads to the destruction of its own cyclin. The noncyclin part of MPF, the Cdk, persists in the cell, inactive until it becomes part of MPF again by associating with new cyclin molecules synthesized during the S and G2 phases of the next round of the cycle.

The fluctuating activities of different cyclin-Cdk complexes are of major importance in controlling all the stages of the cell cycle; they also give the go-ahead signals at some checkpoints. As mentioned above, MPF controls the cell's passage through the G2 checkpoint. Cell behavior at the G1 checkpoint is also regulated by the activity of cyclin-Cdk protein complexes. Animal cells appear to have at least three Cdk proteins and several different cyclins that operate at this checkpoint. Next, let's consider checkpoints in more detail.

Stop and Go Signs: Internal and External Signals at the Checkpoints

Animal cells generally have built-in stop signals that halt the cell cycle at checkpoints until overridden by go-ahead signals. Many signals registered at checkpoints come from cellular surveillance mechanisms inside the cell. These signals report whether crucial cellular processes that should have occurred by that point have in fact been completed correctly and thus whether or not the cell cycle should proceed. Checkpoints also register signals from outside the cell. The signals are transmitted within the cell by signal transduction pathways (see Figure 11.6). Three important checkpoints are those in the G1, G2, and M phases (see Figure 12.15).

For many cells, the G1 checkpoint seems to be the most important. If a cell receives a go-ahead signal at the G1 checkpoint, it will usually complete the G1, S, G2, and M phases and divide. If it does not receive a go-ahead signal at that point, it may exit the cycle, switching into a nondividing state called the **Go**

phase(Figure 12.17a). Most cells of the human body are actually in the G0 phase. As mentioned earlier, mature nerve cells and muscle cells never divide. Other cells, such as liver cells, can be “called back” from the G0 phase to the cell cycle by external cues, such as growth factors released during injury.

Figure 12.17

Two important checkpoints.

Two important checkpoints shown: Part a, G 1 checkpoint and part b, M checkpoint.

Part a. G 1 checkpoint. In the absence of a go-ahead signal, a cell exits the cell cycle and enters G 0, a non-dividing state.

- Diagram 1: Cell hits red gate at the G 1 checkpoint and does not pass, moving instead to the G0 phase. If a cell receives a go-ahead signal, the cell continues on in the cell cycle.
- Diagram 2: Cell passes the G 1 checkpoint and continues in the cell cycle.

Part b. M checkpoint. A cell in mitosis receives a stop signal when any of its chromosomes are not attached to spindle fibers.

- Diagram 1: A cell with chromosomes in prometaphase (most chromosomes are nearing the metaphase plate, but some remain distant) corresponds to the red gate at the M checkpoint in the cell cycle. The cell receives a stop signal. When all chromosomes are attached to spindle fibers from both poles, a go-ahead signal allows the cell to proceed into anaphase.
- Diagram 2: The cell with all chromosomes lined up at the metaphase plate corresponds with the M checkpoint in the cell cycle. The cell receives the go-ahead signal and proceeds to anaphase, with sister chromatids separated.

At certain checkpoints in the cell cycle (red gates), cells do different things depending on the signals they receive. Events of the (a) G1 and (**b**) M checkpoints are shown. In (**b**), the G2checkpoint has already been passed by the cell.

WHAT IF? *What might be the result if the cell ignored either checkpoint and progressed through the cell cycle?*

*For suggested answer, see**Appendix A*.

Biologists are currently working out the pathways that link signals originating inside and outside the cell with the responses by cyclin-dependent kinases and other proteins. An example of an internal signal occurs at the third important checkpoint, the M checkpoint **(Figure 12.17b)**. Anaphase, the separation of sister chromatids, does not begin until all the chromosomes are properly attached to the spindle at the metaphase

plate. Researchers have learned that as long as some kinetochores are unattached to spindle microtubules, the sister chromatids remain together, delaying anaphase. Only when the kinetochores of all the chromosomes are properly attached to the spindle does the appropriate regulatory protein complex become activated. (In this case, the regulatory molecule is not a cyclin-Cdk complex but, instead, a different complex made up of several proteins.) Once activated, the complex sets off a chain of molecular events that activates the enzyme separase, which cleaves the cohesins, allowing the sister chromatids to separate. This mechanism ensures that daughter cells do not end up with missing or extra chromosomes.

There are checkpoints in addition to those in G1, G2, and M. For instance, a checkpoint in S phase stops cells with DNA damage from proceeding in the cell cycle. And, in 2014, researchers presented evidence for another checkpoint between anaphase and telophase that ensures anaphase is completed and the chromosomes are well separated before cytokinesis can begin, thus avoiding chromosomal damage.

What about the stop and go-ahead signals themselves—what are the signaling molecules? Studies using animal cells in culture have led to the identification of many external factors, both chemical and physical, that can influence cell division. For example, cells fail to divide if an essential nutrient is lacking in the culture medium. (This is analogous to trying to run a washing machine without the water supply hooked up; an internal sensor won't allow the machine to continue past the point where water is needed.) And even if all other conditions are favorable, most types of mammalian cells divide in culture only if the growth medium includes specific growth factors. As mentioned in Concept 11.1, a **growth factor** is a protein released by certain cells that stimulates other cells to divide. Different cell types respond specifically to different growth factors or combinations of growth factors.

Consider, for example, *platelet-derived growth factor* (*PDGF*), which is made by blood cell fragments called platelets. When an injury occurs, platelets release PDGF in the vicinity. The experiment illustrated in **Figure 12.18** demonstrates that PDGF is required for the division of cultured **fibroblasts**, a type of connective tissue cell. Fibroblasts have PDGF receptors on their plasma membranes. The binding of PDGF molecules to these receptors (which are receptor tyrosine kinases; see Figure 11.8) triggers a signal transduction pathway that allows the cells to pass the G1 checkpoint and divide. PDGF stimulates fibroblast division not only in the artificial conditions of cell culture but also in an animal's body. Thus, injury results in a proliferation of fibroblasts that helps heal the wound.

Figure 12.18

The effect of platelet-derived growth factor (PDGF) on cell division.

With platelet-derived growth factor (P D G F), cells proliferate. Without P D G F, cells fail to divide.

- A sample of human connective tissue is cut up into small pieces in a Petri dish using scalpels.

- Enzymes are used to digest the extracellular matrix in the tissue pieces, resulting in a suspension of free fibroblasts shown in a flask.
- Cells are transferred to culture vessels (rectangular bottle-like containers that can lie flat) containing a basic growth medium consisting of glucose, amino acids, salts, and antibiotics (to prevent bacterial growth).
- P D G F is added to half the vessels. The culture vessels are incubated at 37 degrees Celsius for 24 hours. Without P D G F (the control) in the basic growth medium, the cells fail to divide and few are shown in the culture vessel. With P D G F, in the basic growth medium, the cells proliferate. Many are shown in the culture vessel. An S E M shows cultured fibroblasts (scale: 10 micrometers).

MAKE CONNECTIONS *PDGF signals cells by binding to a cell-surface receptor tyrosine kinase. If you added a chemical that blocked phosphorylation, how would the results differ? (See*Figure 11.8*.)**

*For suggested answer, see*Appendix A*.**

The effect of an external physical factor on cell division is clearly seen in **density-dependent inhibition**, a phenomenon in which crowded cells stop dividing **(Figure 12.19a)**. Studies done many years ago showed that cultured cells normally divide until they form a single layer of cells on the inner surface of a culture flask, at which point the cells stop dividing. If some cells are removed, those bordering the open space begin dividing again and continue until the vacancy is filled. Follow-up studies revealed that the binding of a cell-surface protein to its counterpart on an adjoining cell sends a signal to both cells that inhibits cell division, preventing them from moving forward in the cell cycle, even in the presence of growth factors.

Figure 12.19

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    Density-dependent inhibition and anchorage dependence of
cell division.
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Density-dependent inhibition and anchorage dependence of cell division.

Individual cells are shown disproportionately large in the drawings.

Part a. Normal mammalian cells. Cell density is limited to a single layer by contact with neighboring cells and the availability of nutrients, growth factors, and a substratum for attachment.

First culture vessel: Cells anchor to the dish surface and divide (anchorage dependence). Few cells are shown.

Second culture vessel: When cells have formed a complete single layer, they stop dividing (density-dependent inhibition). Cells pictured in the culture vessel drawing take up only a single layer.

Third culture vessel: If some cells are scraped away, the remaining cells divide to fill the gap and then stop once they contact each other (density-dependent inhibition). The drawing shows a small patch of cells being scraped away inside the culture vessel.

Fourth culture vessel: A single layer of cells again fills the vessel. The image zooms in on cells. Scale: 20 micrometers.

Part b. Cancer cells. Cancer cells usually continue to divide well beyond a single layer, forming a clump of overlapping cells. They do not exhibit anchorage dependence or density-dependent inhibition.

Most animal cells also exhibit **anchorage dependence** (see Figure 12.19a). To divide, they must be attached to something, such as the inside of a culture flask or the extracellular matrix of a tissue. Experiments suggest that, like cell density, anchorage is signaled to the cell cycle control system via pathways involving plasma membrane proteins and elements of the cytoskeleton linked to them.

Density-dependent inhibition and anchorage dependence appear to function not only in cell culture but also in the body's tissues, checking the growth of cells at some optimal density and location during embryonic development and throughout an organism's life. Cancer cells, which we examine next, exhibit neither density-dependent inhibition nor anchorage dependence **(Figure 12.19b)**.

Loss of Cell Cycle Controls in Cancer Cells

Cancer cells do not heed the normal signals that regulate the cell cycle. In culture, they do not stop dividing when growth factors are depleted. A logical hypothesis is that cancer cells do not need growth factors in their culture medium to grow and divide. They may make a required growth factor themselves, or they may have an abnormality in the signaling pathway that conveys the growth factor's signal to the cell cycle control system even in the absence of that factor. Another possibility is an abnormal cell cycle control system. In these scenarios, the underlying basis of the abnormality is almost always a change in one or more genes (for example, a mutation) that alters the function of their protein products, resulting in faulty cell cycle control. We'll explore the molecular basis for such changes in Concept 18.5.

There are other important differences between normal cells and cancer cells that reflect malfunctions of the cell cycle. If and when they stop dividing, cancer cells do so at random points in the cycle, rather than at the normal checkpoints. Moreover, cancer cells can go on dividing indefinitely in culture if they are given a continual supply of nutrients; in essence, they are "immortal." A striking example is a cell line that has been reproducing in culture since 1951. Cells of this line are called HeLa cells because their original source was a tumor removed from a woman named Henrietta Lacks. (Neither Ms. Lacks nor her family gave permission

or even knew about the propagation and use of her cells, which have helped biologists make countless significant discoveries over the years.) Cells in culture that acquire the ability to divide indefinitely are said to have undergone a process called **transformation**, causing them to behave (in cell division, at least) like cancer cells. By contrast, nearly all normal, nontransformed mammalian cells growing in culture divide only about 20 to 50 times before they stop dividing, age, and die. Finally, cancer cells evade the normal controls that trigger a cell to undergo apoptosis when something is wrong—for example, when an irreparable mistake has occurred during DNA replication preceding mitosis.

Abnormal cell behavior in the body can be catastrophic. The problem begins when a single cell in a tissue undergoes the first of many steps that convert a normal cell to a cancer cell. Such a cell often has altered proteins on its surface, and the body's immune system normally recognizes the cell as "nonself"—an insurgent—and destroys it. However, if the cell evades destruction, it may proliferate and form a tumor, a mass of abnormal cells within otherwise normal tissue. The abnormal cells may remain at the original site if their genetic and cellular changes don't allow them to move to or survive at another site. In that case, the tumor is called a **benign tumor**. Most benign tumors do not cause serious problems (depending on their location) and can be removed by surgery. In contrast, a **malignant tumor** includes cells whose genetic and cellular changes enable them to spread to new tissues and impair the functions of one or more organs; these cells are also sometimes called *transformed* cells (although usage of this term is generally restricted to cells in culture). An individual with a malignant tumor is said to have cancer **(Figure 12.20)**.

Figure 12.20

The growth and metastasis of a malignant breast tumor.

The growth and metastasis of a malignant breast tumor.

A series of genetic and cellular changes contribute to a tumor becoming malignant (cancerous). The cells of malignant tumors grow in an uncontrolled way and can spread to neighboring tissues and, via lymph and blood vessels, to other parts of the body (metastasis).

- A tumor grows from a single cancer cell. The tumor is pictured as a mass of cells in a drawing of a human breast with glandular tissue labeled.
- Cancer cells invade neighboring tissue. The mass of cells has grown and now invades glandular tissue.
- Cancer cells spread through lymph and blood vessels to other parts of the body. Tumor cells leave the breast tissue and travel in both lymph and blood vessels to other parts of the body.
- A small percentage of cancer cells may metastasize to another part of the body. The drawing shows a metastatic tumor growing in a distant part of the body as a result of cancer cells traveling via the

blood and lymph vessels. Breast cancer cells are shown in a colorized S E M (scale: 5 micrometers).

HHMI BioInteractive Animation

How Tumors Grow

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

The changes that have occurred in cells of malignant tumors show up in many ways besides excessive proliferation. These cells may have unusual numbers of chromosomes, though whether this is a cause or an effect of tumor-related changes is an ongoing debate. Their metabolism may be altered, and they may cease to function in any constructive way. Abnormal changes on the cell surface cause cancer cells to lose attachments to neighboring cells and the extracellular matrix, allowing them to spread into nearby tissues. Cancer cells may also secrete signaling molecules that cause blood vessels to grow toward the tumor. A few tumor cells may separate from the original tumor, enter blood vessels and lymph vessels, and travel to other parts of the body. There, they may proliferate and form a new tumor. This spread of cancer cells to locations distant from their original site is called **metastasis** (see Figure 12.20).

A tumor that appears to be localized may be treated with high-energy radiation, which damages DNA in cancer cells much more than DNA in normal cells, apparently because the majority of cancer cells have lost the ability to repair such damage. To treat known or suspected metastatic tumors, chemotherapy is used, in which drugs that are toxic to actively dividing cells are administered through the circulatory system. As you might expect, chemotherapeutic drugs interfere with specific steps in the cell cycle. For example, the drug Taxol freezes the mitotic spindle by preventing microtubule depolymerization, which stops actively dividing cells from proceeding past metaphase and leads to their destruction. The side effects of chemotherapy are due to the effects of the drugs on normal cells that divide frequently, due to the function of that cell type in the organism. For example, nausea results from chemotherapy's effects on intestinal cells, hair loss from effects on hair follicle cells, and susceptibility to infection from effects on immune system cells. You can work with data from an experiment involving a potential chemotherapeutic agent in the **Scientific Skills Exercise**.

Scientific Skills Exercise

Interpreting Histograms

AP Science Practices 1, 2, 4, 6

At What Phase Is the Cell Cycle Arrested by an Inhibitor? Many medical treatments are aimed at stopping cancer cell proliferation by blocking the cell cycle of cancerous tumor cells. One potential treatment is a cell cycle inhibitor derived from human umbilical cord stem cells. In this exercise, you will compare two histograms to determine where in the cell cycle the inhibitor blocks the division of cancer cells.

Human glioblastoma glows green.

Human brain cancer cell

How the Experiment Was Done In the treated sample, human glioblastoma (brain cancer) cells were grown in tissue culture in the presence of the inhibitor, while a control sample of glioblastoma cells was grown in its absence. After 72 hours of growth, the two cell samples were harvested. To get a "snapshot" of the phase of the cell cycle each cell was in at that time, the samples were treated with a fluorescent chemical that binds to DNA and then run through a flow cytometer, an instrument that records the fluorescence level of each cell. Computer software then graphed the number of cells in each sample with a particular fluorescence level, as shown below.

Data from the Experiment

Histograms showing the y-axis has number of cells from 0 to 200 in increments of 40 units and the x-axis has amount of fluorescence per cell (fluorescence units) from 0 to 600 in increments of 200 units, for the Control and the Treated samples.

In each case the histograms are broken into 3 sections. For the Control samples, section A starts at 150, 10, peaks at 200, 130 and ends at 250, 20; section B starts at 250, 20, drops to 275, 15, and ends at 325, 20; section C starts at 325, 20, peaks at 375, 40 and drops to 425, 5. For the Treated samples, section A starts at 150, 0, peaks at 200, 170, and ends at 250, 20; section B starts at 250, 20, drops to 275, 10, and ends at 325, 20; section C starts at 325, 20 and drops to 425, 5.

Data from K. K. Velpula et al., Regulation of glioblastoma progression by cord blood stem cells is mediated by downregulation of cyclin D1, *PLoS ONE* 6(3):e18017 (2011).

The data are plotted in a type of graph called a histogram (above), which groups the values for a numeric variable on the x-axis into intervals. A histogram allows you to see how all the experimental subjects (cells,

in this case) are distributed along a continuous variable (amount of fluorescence). In these histograms, the bars are so narrow that the data appear to follow a curve for which you can detect peaks and dips. Each narrow bar represents the number of cells observed to have a level of fluorescence in the range of that interval. This in turn indicates the relative amount of DNA in those cells. Overall, comparing the two histograms allows you to see how the DNA content of this cell population is altered by the treatment.

INTERPRET THE DATA

- Study the data in the histograms.
- Which axis indirectly shows the relative amount of DNA per cell? Explain your answer.
- In the control sample, compare the first peak in the histogram (in region A) to the second peak (in region C). Which peak shows the population of cells with the higher amount of DNA per cell? Explain. (For additional information about histograms, see the Scientific Skills Review in Appendix D.)
-
    - In the control sample histogram, identify the phase of the cell cycle (G1, S, or G2) of the population of cells in each region delineated by vertical lines. Label the histogram with these phases and explain your answer.
- Does the S phase population of cells show a distinct peak in the histogram? Why or why not?
- The histogram representing the treated sample shows the effect of growing the cancer cells alongside human umbilical cord stem cells that produce the potential inhibitor.
- Label the histogram with the cell cycle phases. Which phase of the cell cycle has the greatest number of cells in the treated sample? Explain.
- Compare the distribution of cells among G1, S, and G2 phases in the control and treated samples. What does this tell you about the cells in the treated sample?
- Based on what you learned in Concept 12.3, propose a mechanism by which the stem cell–derived inhibitor might arrest the cancer cell cycle at this stage. (More than one answer is possible.)

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Over the past several decades, researchers have produced a flood of valuable information about cell-signaling pathways and how their malfunction contributes to the development of cancer through effects on the cell cycle. Coupled with new molecular techniques, such as the ability to rapidly sequence the DNA of cells in a particular tumor, medical treatments for cancer are beginning to become more “personalized” to a particular patient’s tumor.

BBC Video

Are Fruit Flies the Key in the Fight Against Cancer?

image

Play

image

Play

For example, the cells of roughly 20% of breast cancer tumors show abnormally high amounts of a cell-surface receptor tyrosine kinase called HER2, and many show an increase in the number of estrogen receptor (ER) molecules, intracellular receptors that can trigger cell division. Based on lab findings, a physician can prescribe chemotherapy with a molecule that blocks the function of the specific protein (Herceptin for HER2 and tamoxifen for ERs). Treatment using these agents, when appropriate, has led to increased survival rates and fewer cancer recurrences (see Make Connections Figure 18.27).

Interview

Interview with Bruce Alberts: Cancer control and careers in science

A photo of Bruce Alberts.